Linux/macOS `uname` Command Notes

The `uname` command is used to display **system information** such as kernel name, version, architecture, and hostname.

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1. `uname` (or `uname -s`)

Meaning: Prints the **kernel name**.

- **Linux Example:**

```
$ uname
Linux
```

- **macOS Example:**

```
$ uname
Darwin
```

👉Shows whether you are running Linux (`Linux`) or macOS (`Darwin`).

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2. `uname -r`

Meaning: Prints the **kernel release version**.

- **Linux Example:**

```
$ uname -r
5.15.0-58-generic
```

- **macOS Example:**

```
$ uname -r
23.0.0
```

👉Useful for checking your **kernel release** (e.g., to match drivers or kernel modules).

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3. `uname -v`

Meaning: Prints the **kernel build version** (extra build info).

- **Linux Example:**

```
$ uname -v
#64-Ubuntu SMP Thu Jan 5 11:00:00 UTC 2023
```

- **macOS Example:**

```
$ uname -v
Darwin Kernel Version 23.0.0: Mon Aug 14 2023; root:xnu-10002.1.13~1/
RELEASE_ARM64_T8103
```

👉Shows **detailed build info**, often used by developers or sysadmins.

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4. `uname -m`

Meaning: Prints the **machine hardware name (architecture)**.

- **Linux Example:**

```
$ uname -m
x86_64
```

- **macOS Example (Apple Silicon):**

```
$ uname -m
arm64
```

👉 Useful to know whether your system is **64-bit (x86_64)** or **ARM-based (arm64, M1/M2/M3/M4 Macs)**.

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5. `uname -p`

Meaning: Prints the **processor type** (may show `unknown` on some systems).

- **Linux Example:**

```
$ uname -p
x86_64
```

- **macOS Example:**

```
$ uname -p
arm
```

👉Sometimes unreliable. Prefer `uname -m` for architecture.

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6. `uname -i`

Meaning: Prints the **hardware platform** (may be unknown).

- **Linux Example:**

```
$ uname -i
x86_64
```

- **macOS Example:**

```
$ uname -i
unknown
```

👉Works mostly on Linux, not reliable on macOS.

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7. `uname -o`

Meaning: Prints the **operating system** (Linux only).

- **Linux Example:**

```
$ uname -o
GNU/Linux
```

- **macOS Example:**

```
$ uname -o
uname: illegal option -- o
usage: uname [-amnprsv]
```

👉**Not supported on macOS.**

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8. `uname -n`

Meaning: Prints the **node name (hostname)** of your system.

- **Linux Example:**

```
$ uname -n
prashant-laptop
```

- **macOS Example:**

```
$ uname -n
MacBook-Pro.local
```

👉When to use: - To check your **system hostname** quickly. - Useful when working with **networks, servers, and SSH sessions** to know which machine you are on.

😯Alternative commands for hostname:

```
$ hostname
$ cat /etc/hostname   # Linux only
```

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9. `uname -a`

Meaning: Prints **all system information** in one line.

- **Linux Example:**

```
$ uname -a
Linux prashant 5.15.0-58-generic #64-Ubuntu SMP Thu Jan 5 11:00:00 UTC
2023 x86_64 x86_64 x86_64 GNU/Linux
```

- **macOS Example:**

```
$ uname -a
Darwin MacBook-Pro.local 23.0.0 Darwin Kernel Version 23.0.0: Mon Aug
14 2023; root:xnu-10002.1.13~1/RELEASE_ARM64_T8103 arm64
```

👉Best for a **quick overview**: kernel, version, architecture, hostname, OS.

Summary Table

Command	Meaning	Linux Example	macOS Example
`uname`	Kernel name	Linux	Darwin
`uname -s`	Same as `uname` (kernel name)	Linux	Darwin
`uname -r`	Kernel release version	5.15.0-58-generic	23.0.0
`uname -v`	Kernel version (build info)	#64-Ubuntu SMP ...	Darwin Kernel Version 23.0.0
`uname -m`	Machine hardware (architecture)	x86_64	arm64
`uname -p`	Processor type	x86_64	arm
`uname -i`	Hardware platform	x86_64	unknown
`uname -o`	Operating system	GNU/Linux	👈Not supported
`uname -n`	Node name (hostname)	prashant-laptop	MacBook-Pro.local
`uname -a`	All system information	Linux prashant 5.15...	Darwin MacBook-Pro...

👈 `uname` is a **system info command**: - Developers use it for checking architecture before installing software. - Sysadmins use it to identify kernel version and hostname on servers. - `uname -a` is the quickest way to get a **full snapshot** of system details.